

Mathematics, 2018 EC, entrance exam, version 5

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 51

Total marks: 51

Name _____ Grade _____ ID No. _____ Date _____

1. Which term of the following arithmetic sequence -6, 1, 8, 15, ... is 414? (1)

A. 31

B. 62

C. 61

D. 32
2. $\sum_{n=4}^{17} \frac{1}{(k^2+5k+6)}$ is equal to: (Note: the index likely n, but given as k) (1)

A. 11/102

B. 7/60

C. 12/85

D. 10/119
3. Which of the following Geometric sequence is divergent? (1)

A. $a_n = (-1)^{2n+1}(3/7)^n$

B. $a_n = (-3/5)^{2n+1}$

C. $a_n = (1/10)(5/2)^n$

D. $a_n = 2/5^n$
4. An auditorium has 30 seats in first row, 34 in second, 38 in third. If pattern continues, total seating capacity with 20 rows? (1)

A. 106

B. 3880

C. 580

D. 1360
5. A manufacturer wants to design an open box with square base and surface area 75 square units. What height gives maximum volume? (1)

A. 5 units

B. 2.5 units

C. 7.5 units

D. 15 units
6. What is area of region enclosed between $f(x)=x^2-4$ over [1,5]? (1)

A. 5/3

B. 65/3

C. 16/3

D. 86/3
7. For 10 observations, sum was 120 and sum of squares 1640. If one observation 10 is corrected to 20, then corrected standard deviation is: (1)

A. 5

B. 4

C. 6

D. 7
8. What is the coefficient of quartile range for data: 23,15,8,7,13,24,6,20,26,5? (1)

A. 7/23

B. 7/30

C. 30/23

D. 16/30
9. The mean and standard deviation of incomes: Company A: mean=8000, $\sigma=160$; Company B: mean=5000, $\sigma=120$. Which is true? (1)

A. Company A has more variable income than B.

B. Company B has more consistent income than A.

C. It is not possible to compare variability.

D. Company B has more variable data than A.

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10. A researcher wants proportional representation from different colleges. Which sampling technique is most appropriate? (1)
- A. Simple random sampling
B. Systematic sampling
C. Stratified random sampling
D. Convenience sampling
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11. Solution set of inequality $x/(x-3) \leq 2$ is: (1)
- A. (3,6]
B. $(-\infty, 3) \cup [6, \infty)$
C. $(-\infty, 3) \cup (6, \infty)$
D. [6, ∞)
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12. A dairy farmer wants to minimize cost while meeting requirements. Feed A costs 14/kg, contains 0.4kg protein, 0.2kg fiber. Feed B costs 21/kg, contains 0.3kg protein, 0.5kg fiber. Daily requirement: at least 12kg protein, at least 10kg fiber. Minimum cost? (1)
- A. 450
B. 540
C. 700
D. 840
-
13. A bed sold for 45,000 Birr including 25% markup based on cost. Original cost? (1)
- A. 33,750
B. 36,000
C. 37,500
D. 40,000
-
14. Which is true about an amortization schedule? (1)
- A. Only the total interest paid over the loan
B. A breakdown of each payment into principal, interest, and remaining balance.
C. The bank's profit from the loan
D. Only down payment of the principal
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15. *Tax table provided.* (1)
A shop is rented at Birr 2,800 per month for one year. Tax rates given. What is tax payable?
- A. Birr 3,330
B. Birr 5,040
C. Birr 0
D. Birr 2,800
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16. What is the range of $f(x)=3 + \operatorname{sgn}(x^2+1)$? (1)
- A. The set of all integers
B. {2,3,4}
C. {4}
D. {-1,0,1}
-
17. If $f(x)=ax+b$ and $f^{-1}(2x)=x-3$, then $a=?$ (1)
- A. 3
B. 1
C. -3
D. 2
-
18. What are values A, B, C so that $(4x^2 - x + 2)/(x^3 + x) = A/x + (Bx+C)/(x^2+1)$? (1)
- A. -1,1,1
B. -1,1,2
C. -1,-2,0
D. 2,2,-1
-
19. Which is true about asymptote(s) of $f(x)=(x^2-5x+6)/[x(x-3)(x-1)]$? (1)
- A. It has an oblique asymptote.
B. The graph does not intersect its asymptotes.
C. As $x \rightarrow 3^+$, $f(x) \rightarrow +\infty$
D. As $x \rightarrow 0^-$, $f(x) \rightarrow -\infty$

20. Suppose A and B are 3×3 matrices, I identity, such that $AB=3I$ and $\det(A)=9$. Then $\det(B)=?$ (1)
- A. $1/3$ B. 3
C. 27 D. $1/9$
-
21. Consider system: $x+2y-z=0$, $2x+5y+z=0$, $3x+by-2z=1$. Determinant of coefficient matrix is 8. Find x and y? (1)
- A. $7/8$ and $3/8$ B. $5/8$ and $-5/8$
C. $7/8$ and $-3/8$ D. $-3/8$ and 1
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22. If vectors $u=(3,-2)$, $v=(-2,1)$, $w=(3,-3)$, find a such that $a + w = 2u - v$. (1)
- A. (5,2) B. (5,-2)
C. (11,2) D. (-2,5)
-
23. Let $u=(5,2)$, $v=(2,-2)$. Then $|2u - v|$ is: (1)
- A. -8 B. 3
C. 24 D. 10
-
24. What is area of parallelogram spanned by vectors $i+2j+k$ and $2i - j + 2k$? (1)
- A. $5\sqrt{2}$ B. 5
C. $2\sqrt{5}$ D. $2\sqrt{19}$
-
25. What is image of circle $x^2+y^2-6x+4y+12=0$ under translation $T(2,-3)$? (1)
- A. $x^2+y^2-10x+10y+49=0$ B. $x^2+y^2+10x-6y+49=0$
C. $x^2+y^2-6x+8y+9=0$ D. $x^2+y^2-10x-10y-1=0$
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26. What is image of line $y=2x+1$ after reflecting along line $y=x-3$? (1)
- A. $x+2y+10=0$ B. $2x-y-10=0$
C. $x+y-10=0$ D. $x-2y-10=0$
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27. If modal value of data is 67.5, find m. (Table: 41-50:4, 51-60:2, 61-70:m, 71-80:6, 81-90:1, 91-100:8) (1)
- A. 7 B. 8
C. 9 D. 10
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28. In how many ways can a committee of 4 be selected from 6 boys and 4 girls if committee contains at most 2 boys? (1)
- A. 85 B. 115
C. 24 D. 125
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29. Which is true about events $E_1=\{2,3,5\}$, $E_2=\{1,6\}$, $E_3=\{2,4,6\}$ for rolling a die? (1)
- A. They are mutually exclusive. B. They are exhaustive events.
C. $P(E_1 \cup E_3)=1$ D. $P(E_1 \cap E_3)=1/4$
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30. A committee of 3 from 10 candidates including 4 girls. Probability that at most one girl is selected? (1)
- A. $2/3$ B. $3/4$
C. $1/3$ D. $5/6$
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31. Which of the following is onto function? (1)
- A. $f:R \rightarrow R$, $f(x)=|x|$ B. $f:R \rightarrow R$, $f(x)=x^2$
C. $f:R \rightarrow [0, \infty)$, $f(x)=3^x$ D. $f:(0, \infty) \rightarrow R$, $f(x)=\log_2 x$

32. Given $f=\{(1,0),(-1,2),(3,4),(2,1)\}$ and $g=\{(0,0),(-1,2),(3,4),(2,1)\}$. Which is correct? (1)
- A. $2f(2)=2g(1)$ B. $(f+g)(-1)=1$
C. $(f-g)(3)=0$ D. $(f\cdot g)(2)=-1$
33. Which is NOT true about $f(x)=2x^5-7x^3+2x-1$? (1)
- A. It has at most 4 turning points and intersects x-axis at most 5 times.
B. As we move far from origin it rises to right and falls to left.
C. It is a smooth and continuous curve.
D. It passes through the origin.
34. What is value of b such that $x-1$ and $x+1$ are factors of $ax^3 - bx^2 + 3x - 10$? (1)
- A. -10 B. -7
C. 7 D. 14
35. Simplified form of $3^{2+\log_9 25}$ is: (1)
- A. 75 B. 45
C. 9 D. 5
36. Which one is false about $f(x)=\log_3(x-3)$? (1)
- A. $f(x)\geq 0$ for $x\geq 3$ B. $0<f(x)<1$ for $4<x<6$
C. Domain is real numbers greater than 3 D. Line $x=3$ is vertical asymptote
37. What is $\sec(240^\circ)$? (1)
- A. $-2\sqrt{3}/3$ B. 2
C. -2 D. $2\sqrt{3}/3$
38. If θ is second quadrant angle and $\csc\theta = 5/3$, then $\cot\theta = ?$ (1)
- A. $4/5$ B. $3/4$
C. $4/3$ D. $-4/3$
39. A regular pyramid has square base side 10 cm, lateral edge 13 cm. Total surface area? (1)
- A. 240 cm^2 B. 340 cm^2
C. 260 cm^2 D. 360 cm^2
40. What is point that divides segment P(2,3) and Q(-2,4) internally in ratio 1:2? (1)
- A. $(2/3, -10/3)$ B. $(10/3, 5/3)$
C. $(2/3, 10/3)$ D. $(-2/3, 11/3)$
41. Out of 135 students, 59 take Math, 99 take English, 10 take neither. How many take both? (1)
- A. 43 B. 59
C. 33 D. 23
42. If $x=2.131131113\dots$ and $y=7.868868886\dots$, find $(x+y)/5$? (1)
- A. 20 B. 2
C. 1 D. $1/5$

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- 43.** Which is not true about real numbers? (1)
- A. Difference of two irrational numbers is always irrational.
 - B. Product of nonzero rational and irrational is irrational.
 - C. Between any two irrationals there is a rational.
 - D. If $a < b$ then $a < (a+b)/2 < b$
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- 44.** The solution of $|2x-4|+|x+5|=8$ is: (1)
- A. $\{-3,1\}$
 - B. $\{1,7/3\}$
 - C. $\{-3,1,7/3\}$
 - D. $\{-3,7/3\}$
-
- 45.** A building casts shadow 45 m when angle of elevation of sun is 30° . Height of building? (1)
- A. $45\sqrt{3}$
 - B. $45\sqrt{3}/2$
 - C. $45\sqrt{2}/2$
 - D. $45\sqrt{3}/3$
-
- 46.** If each exterior angle of a regular polygon is 40° , how many sides? (1)
- A. 8
 - B. 10
 - C. 7
 - D. 9
-
- 47.** Area of triangle 128 cm², one side 8 cm, corresponding side of similar triangle 12 cm. Area of other triangle? (1)
- A. 129
 - B. 192
 - C. 288
 - D. 64
-
- 48.** Let $u=(5,3)$, $v=(-8,1)$. Unit vector in direction of $u+v$? (1)
- A. $3/5 i - 2/5 j$
 - B. $-3/5 i + 4/5 j$
 - C. $1/2 i - 2/5 j$
 - D. $4/5 i + 2/5 i$
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- 49.** Mean salary of 10 employees = 6450. New employee gets 7000. New mean? (1)
- A. 6650
 - B. 6600
 - C. 6550
 - D. 6500
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- 50.** If variance of a,b,c,d is 9, then standard deviation of $2a+3, 2b+3, 2c+3, 2d+3$? (1)
- A. 36
 - B. 18
 - C. 9
 - D. 6
-
- 51.** Two dice thrown. Probability sum multiple of 5 or greater than 9? (1)
- A. $5/18$
 - B. $1/4$
 - C. $11/36$
 - D. $13/36$

— END OF EXAMINATION —